

****Establishment and Sharing of Datasets****

****Date: 30-01-2024****

****Decision****

****Private Companies****

****Response to Inquiry****

****Legal Basis for Processing****

****Biometrics****

The Data Protection Agency has received an inquiry from the Alexandra Institute, which has requested the Agency's assessment of the possibility of establishing and sharing a dataset intended for the development of Danish language technology.

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****Summary****

The Data Protection Agency has received an inquiry from the Alexandra Institute, which has requested the Agency's assessment of the possibility of establishing and sharing a dataset intended for the development of Danish language technology. In this context, the Agency has addressed questions regarding the legal basis for processing in relation to the establishment and sharing of such datasets, anonymization, and whether the processing involves special categories of personal data (link: <https://www.datatilsynet.dk/hvad-siger-reglerne/grundlaeggende-begreber/hvad-er-personoplysninger>) when processing personal data in the form of a person's voice. The Data Protection Agency has assessed that the collection and processing of personal data as part of the establishment and publication of the dataset can occur with reference to the fulfillment of a contract (link: <https://www.datatilsynet.dk/hvad-siger-reglerne/grundlaeggende-begreber/hvornaar-maa-du-behandle-personoplysninger>).

Furthermore, the Data Protection Agency has assessed that the dataset cannot be considered anonymized in the data protection legal sense. This is due to the existence of tools that could reasonably be used to identify the individuals represented in the dataset. Finally, the Data Protection Agency has determined that the processing—despite potentially involving biometric data—is not covered by the prohibition against processing special categories of personal data under GDPR. This is because the Alexandra Institute's processing of the information does not aim to unambiguously identify a physical person.

You can read the Data Protection Agency's statement to the Alexandra Institute below.

1. **The Inquiry**

On behalf of the Alexandra Institute, you submitted an inquiry to the Data Protection Agency on June 21, 2023, regarding the possibility of establishing and sharing a dataset intended for the development of Danish speech technology. The establishment of the dataset will involve individuals of various genders, ages, and geographical backgrounds reading and recording Danish texts.

You specifically asked whether this situation would involve the processing of special categories of personal data under Article 9 of the General Data Protection Regulation, as well as whether there are other aspects regarding the collection and processing of the dataset that the Data Protection Agency finds necessary to comment on. According to the Data Protection Agency, the inquiry raises questions about the legal basis for processing in relation to the establishment and sharing of such datasets, as well as about anonymization.

2. **The Data Protection Agency's Assessment**

After the case has been presented to the Data Council, the Data Protection Agency assesses that the collection and processing of personal data as part of the establishment and publication of the dataset can occur based on the legal basis in Article 6(1)(b) of the General Data Protection Regulation concerning the fulfillment of a contract.

Furthermore, the Data Protection Agency assesses that the audio recordings—also after the dataset has been cleansed of any metadata—constitute personal data in this specific case, and therefore the

data protection rules apply. Finally, the Data Protection Agency assesses that the information in the dataset may constitute biometric data. However, the processing of the information will not be covered by the prohibition against processing special categories of personal data in Article 9 of the General Data Protection Regulation, as the processing will not aim to unambiguously identify a person.

A justification for the Data Protection Agency's assessment is provided below.

3. ****Background for the Assessment****

3.1. ****About the Project****

The Alexandra Institute is participating in a project aimed at developing and improving Danish speech technology. The project also involves two private companies, the Department of Computer Science at the University of Copenhagen, and the Agency for Digitization.

As part of the project, a speech dataset named CoRal, which stands for Danish Conversational and Read-Aloud Speech Dataset, will be developed over the next two years. The dataset is intended to contain 1000-1500 hours of conversation and read-aloud speech from a broad and representative sample of the population concerning gender, age, Danish dialects, and foreign accents. At the same time, language models will be developed that can recognize Danish speech and read Danish text aloud. All data and models will be tested and published continuously, so third parties, including developers, companies, and public institutions, can benefit from them.

The ambition is to recruit more than 2000 individuals of various genders, ages, and geographical backgrounds to read and record Danish texts, thereby creating the dataset. The recordings will be made based on a written agreement with each reader and the Alexandra Institute. The agreement will include information about the citizen's name, age, current postal code, postal code at birth, dialect, email address, and phone number. The text that...

****3.1. Data Protection and Anonymization****

The recordings will not contain personal data. However, the audio files will contain the individual's voice. The dataset will be cleaned so that information that is directly identifiable, such as name and address, will be deleted. The individual audio files will, however, be associated with general information about the speaker, including gender, age, and approximate geographical location (e.g., female, 60-70 years, East Jutland). If the number of speakers in a group, such as an age group, is few, the information will not be included in the dataset to avoid potential identification of individuals.

****3.2. Legal Basis for Processing****

The question is primarily whether the Alexandra Institute can process the information in the form of audio recordings of individuals who voluntarily sign up to record Danish texts, based on Article 6, paragraph 1, letter b of the General Data Protection Regulation (GDPR), concerning the fulfillment of a contract. Article 6, paragraph 1, letter b of the GDPR allows organizations to process information about a person when it is necessary to fulfill a contract with that person.

The Court of Justice of the European Union (CJEU) addressed the question of when the processing of personal data can be considered necessary for the fulfillment of a contract in its judgment of July 4, 2023, in case C-252/21, Meta Platforms et al. According to the CJEU, the processing of personal data must be "objectively necessary to achieve a purpose that is an integral part of the contractual service to the data subject. The data controller must therefore be able to demonstrate how the main purpose of the contract cannot be fulfilled without the processing in question."

In this case, the contract concerns the reading and recording of a number of texts. The service that the data subject must provide is one or more audio recordings, in which the individual's voice will particularly be evident. The audio recordings will subsequently be released under a license that permits anyone to copy and redistribute the material. In return, the data subject receives a gift card of a specified amount.

On this basis, the Danish Data Protection Agency assesses that Article 6, paragraph 1, letter b of the GDPR can serve as the basis for the processing activity concerning the reading and recording of

Danish texts. The Agency particularly emphasizes that the recording of the relevant audio recordings is the primary contractual service that the data subject must provide and constitutes the main purpose of the contract. Thus, the main purpose of the contract cannot be fulfilled without the recording of the audio recordings. The Agency assumes in this context that the information contained in the contract itself, such as name, email, phone number, etc., is also necessary for the fulfillment of the contract, e.g., delivery of the gift card.

****3.3. Is it a Matter of Anonymization?****

The inquiry also raises questions about whether there is real anonymization when the recordings are otherwise cleansed of all direct identifiers, and the recording itself does not otherwise contain personal data, or whether audio recordings with a person's voice should be considered personal data in themselves.

****3.1.1. General Data Protection Regulation, etc.****

Article 4, number 1 of the GDPR defines the concept of personal data as any information relating to an identified or identifiable natural person. An identifiable natural person is understood as a natural person who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, identification number, location data, an online identifier, or one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person.

Recital 26 of the GDPR states the following about the concept of personal data: "Personal data that has been subject to pseudonymization and that can be attributed to a natural person by the use of additional information should be considered to be information about an identifiable natural person. To determine whether a natural person is identifiable, all means should be considered that may reasonably be used by the data controller or another person to identify the individual directly or indirectly, including by reference to that individual. To ascertain whether means may reasonably be used to identify a natural person, all objective factors should be taken into account, such as the costs of and time required for identification, taking into account the available technology at the time of processing and the technological developments.

The principles of data protection should therefore not apply to anonymous information, i.e., information that does not relate to an identified or identifiable natural person, or to personal data that has been made anonymous in such a way that the data subject is not or no longer identifiable." (Emphasis added by the Danish Data Protection Agency)

From the Ministry of Justice's report no. 1565/2017, p. 44, it appears: "From the remarks to Section 3, number 1 of the Data Protection Act, it is stated, among other things, that the term identifiable person refers to a person who can be identified, directly or indirectly, including by an identification number or one or more elements that are specific to a given person's physical, physiological, mental, economic, cultural, or social identity. The concept of personal data includes, according to...

****Remarks on the following information that can be attributed to a physical person, even if this requires knowledge of personal identification numbers, registration numbers, or similar specific identifiers such as, for example, a serial number. Included will also be, among other things, information that exists in the form of images, the person's voice, fingerprints, or genetic characteristics. It is of no significance whether the identification information is generally known or readily available; therefore, cases where it is only possible for the initiated to understand to whom a piece of information pertains are also included in the definition." (Emphasis by the Data Protection Agency)****

It is noted for the sake of good order that the definition in Article 4, no. 1 of the General Data Protection Regulation and § 3, no. 1 of the Data Protection Act corresponds to the definition of personal data in the previously applicable Data Protection Act § 3, no. 1, and the underlying previous Data Protection Directive Article 2, letter a.

From the Article 29 Working Party's opinion on the concept of personal data [2] (WP 136, no. 4/2007), the following regarding the identification requirement follows (p. 13):

"Generally, a person can be considered 'identified' when that person can be 'distinguished' from all

other members within a group of persons. The physical person is thus 'identifiable' when it is possible to identify that person (this is the meaning of the suffix '-able'), even if it has not yet occurred. This second alternative is therefore, in practice, the boundary condition that determines whether the information falls within the scope of the third element. Identification is usually carried out using specific information, which we can call 'identifiers', and which has a particularly privileged and close connection with the person in question. Examples of this include the person's external characteristics, such as height, hair color, and clothing, or a characteristic of the person that cannot be immediately seen, such as occupation, position, and name. In the directive, these 'identifiers' are mentioned in the definition of personal data in Article 2, where it states that a physical person 'can be identified, directly or indirectly, by an identification number or one or more elements specific to the physical, physiological, psychological, economic, cultural, or social identity of that person.'

***"Directly" or "indirectly" identifiable**

The remarks on the articles in the Commission's amended proposal contain further clarifications, stating that "a person can be identified directly by name or indirectly by a telephone number, a vehicle registration number, a social security number, a passport number, or a combination of significant criteria that makes it possible to recognize him by narrowing down the group to which he belongs (age, occupation, residence, etc.)." This formulation clearly shows that the question of whether certain identifiers are sufficient to ensure identification depends on the circumstances of the specific situation. A very common surname will not be enough to identify a person—i.e., to distinguish that person—when looking at a country's entire population, while it will likely be sufficient to ensure identification of a student in a classroom. Even less significant information, such as "the man in the black suit," can identify a person among passersby at a traffic light. Thus, the question of whether the person to whom the information pertains is identified or not depends on the circumstances of the particular case.

To establish this identity, the person's name sometimes needs to be combined with other information (date of birth, parents' names, address, or a facial image) to avoid confusion between that person and any namesakes. For example, the information that Titius owes a certain amount can be considered to pertain to an identified person because it is associated with that person's name. The name is a piece of information that shows that the person uses that combination of letters and sounds to distinguish himself from others and allows other individuals with whom he establishes connections to differentiate him from others. The name can also serve as a starting point that leads to information about where the person lives or can be found and can also provide information about individuals in his family (through the surname) and a number of different legal and social connections associated with the name (education, medical records, bank accounts). It may even be possible to find out what a person looks like if his image is linked to that name. All these new pieces of information associated with the name can enable a zooming in on a flesh-and-blood person, and through the identifiers, the original information is thus connected with a physical person who can be distinguished from other persons.

Regarding "indirectly" identified or identifiable persons, this category typically concerns the phenomenon of "unique combinations," whether they are small or large. In cases where the scope of the existing identifiers does not initially allow for finding a specific person, that person may still be "identifiable" because this information combined with other information (regardless of whether the record keeper has the latter information or not) will make it possible to distinguish that person from others. Here, the directive comes in with "one or more elements that are specific to the physical, physiological, psychological, economic, cultural, or social identity of that person." Some characteristics are so unique that a person can be identified without problems ("the current Spanish Prime Minister"), but a combination of information at the category level...

(alderskategori, regional oprindelse osv.) can also be quite unambiguous under certain circumstances, especially if one has access to some form of supplementary information. This phenomenon has been closely examined by statisticians, who always strive to avoid breaches of confidentiality.

It should be noted here that a name in itself is not always necessary to identify a person, although identification using the name is the most common method in practice. This can occur when other

"identifiers" are used to pinpoint a person. In computer registers that contain personal data, a unique identifier is usually employed for the registered individuals to avoid confusion between two persons in the register. On the internet, tools for monitoring internet traffic also make it easy to identify a machine's behavior and, behind the machine, the user's behavior. Thus, the individual's personality is pieced together, allowing certain decisions to be attributed to that individual. Without even asking for the person's name and address, it is possible to categorize the individual based on socioeconomic, psychological, philosophical, or other criteria and attribute certain decisions to them, as the individual's contact point (a computer) no longer necessarily requires their identity in a narrow sense to be revealed. In other words, the possibility of identifying a person is no longer necessarily equivalent to the possibility of finding out that person's name. The definition of personal data reflects this relationship.

The Court of Justice of the European Communities has confirmed this, stating that "an operation consisting of referring to different persons on a website and identifying them by name or otherwise, for example, by providing their telephone number or by giving information about their employment and leisure interests, constitutes processing of personal data [...] in the sense in which the term is used in Directive 95/46/EC." (Emphasis added by the Data Protection Authority)

Furthermore, the Court of Justice of the EU, in its judgment of October 19, 2016, in case C-582/14, Breyer, paragraphs 41-49, addressed the question of to what extent and under what circumstances a dynamic IP address can be considered personal data.

The Court stated the following: "The use of the word 'indirect' by the EU legislator seems to indicate that, in order to qualify a piece of information as personal data, it is not necessary for that information in itself to enable the identification of the data subject.

It is also stated in recital 26 of Directive 95/46 that, in order to determine whether a person is identifiable, all means that could reasonably be used to identify that person, either by the data controller or by any other person, are to be taken into account.

Insofar as this recital refers to means that could reasonably be considered to be used, both by the data controller and by 'any other person', it seems to follow from its wording that, for a piece of information to qualify as 'personal data' as referred to in Article 2(a) of that directive, it is not required that all the information enabling the identification of the data subject must be held by a single person.

The fact that the additional knowledge required to identify the user of a website is not held by the provider of online media services, but by that user's internet service provider, does not seem to preclude the possibility that the dynamic IP addresses recorded by the provider of online media services constitute personal data for that provider as referred to in Article 2(a) of Directive 95/46.

However, it must be determined whether the possibility of combining a dynamic IP address with the additional knowledge that the internet service provider has constitutes a means that could reasonably be used to identify the data subject.

As the Advocate General has essentially stated in paragraph 68 of the proposal for a decision, this is not the case if the identification of the data subject is prohibited by law or practically unfeasible, for example, due to the fact that it would involve a greater effort in terms of time, costs, and work, such that the risk of identification seems in reality negligible.

Although the referring court has clarified in its referral decision that German law does not allow the internet service provider to directly transfer the additional knowledge necessary for the identification of the data subject to the provider of online media services, it seems, subject to the verifications that must be carried out by the referring court in this regard, that there are legal avenues that allow the provider of online media services, including in cases of network attacks, to contact the competent authority so that it can take the necessary steps to obtain this information from the internet service provider and to initiate criminal prosecution.

The provider of online media services thus seems to have means that could reasonably be used, with

the assistance of other persons, namely the competent authority and the internet service provider, to identify the data subject based on the stored IP addresses.

In light of all the above considerations, the first question must be answered by stating that Article 2(a) of Directive 95/46 must be interpreted as meaning that a dynamic IP address, which a provider of online media services records in connection with a search conducted by a person on a website, as this provider makes available to the public, in relation to the mentioned provider constitutes a personal data as referred to in this provision, when the provider has legal means that enable it to identify the data subject through the additional knowledge that this person's internet provider possesses." (Data Protection Authority's emphasis)

3.1.2. Assessment

The definition in Article 4, No. 1 of the regulation contains four elements that are part of the assessment of whether it concerns personal data. There must be (i) "any information," (ii) "about," (iii) "identified or identifiable," and (iv) "natural person."

In the present case, the question is primarily whether the requirement of identified or identifiable is met.

"Identified"

It must first be assessed whether a person can be considered identified through their voice alone.

The data protection rules do not contain a more precise specification of what it means for a person to be identified, as follows from the definition in Article 4, No. 1.

According to the Article 29 Working Party, identified should be understood in accordance with the natural understanding of the word, namely that a person is identified "when the individual can be 'distinguished' from all the other members of the group." Identification can, according to the Article 29 Working Party, occur through identifiers, which are information that has a particularly privileged and close connection with the individual, e.g., physical traits.

The Article 29 Working Party refers to a person's name as the most common identifier and that the concept of identified person most often implies a reference to the individual's name. At the same time, the Article 29 Working Party specifies that a name in itself is not always necessary to identify a person. Other identifiers can also be used to pinpoint a person.

The Article 29 Working Party further notes that the question of whether an identifier is sufficiently unique to ensure identification depends on a concrete assessment of the specific situation:

"A very common surname will not be enough to identify a person – i.e., to distinguish the individual – when looking at a country's total population, while it will likely be enough to ensure identification of a student in a classroom. Even less significant information, such as 'the man in the black suit,' can identify a person among the passersby at a traffic light. Thus, the question of whether the person to whom the information relates is identified or not depends on the circumstances of the specific case."

A voice can therefore also constitute an identifier that allows for the identification of a person within a group of people and thus mean that the person is identified. The question is to what extent a person's voice can be considered a sufficiently unique identifier for the person to be directly identified when they appear as part of the dataset that the Alexandra Institute aims to establish.

The Data Protection Authority assumes here that the quality of the audio recording will be high, and therefore it should not be taken into account that the voice on the recording may sound distorted or similar.

Voices have many characteristics and can be unique due to differences in speech patterns, pitch, rhythm, pronunciation, mental state, language, and dialect. A voice can thus be said to be particular to a given person's identity, and these specific characteristics can be used to differentiate one voice from another. Research has further shown that it may be possible to infer information about age, gender, or ethnicity from a person's voice.

However, recognition of the voice must also presuppose prior knowledge of the voice and the person behind it. Friends, family, and other close relations will thus likely be able to identify who the voice belongs to simply by listening to it. For the uninitiated, identification will be more difficult.

In accordance with the Data Protection Authority's practice, the circumstance that certain individuals

can identify the data subject, e.g., based on a description of an event, cannot in isolation lead to the conclusion that the processing of the information must be considered covered by the data protection rules.

However, the Data Protection Authority cannot, based on the information available, with sufficient certainty determine the extent to which it is possible for a broader circle of people to identify a person based on the audio recordings in question, and that the voice therefore already constitutes a unique identifier that leads to the conclusion that the relevant audio recordings must be considered personal data about an identified person.

“Identifiable”

The question then becomes whether the person is indirectly identifiable, i.e., whether there is additional information that makes it possible to pinpoint the individual. This depends on a concrete assessment of the circumstances in the specific situation.

To determine whether a natural person is identifiable, all means that can reasonably be expected to be used by the data controller or another person to directly or indirectly identify the individual should be taken into consideration.

To ascertain whether means can reasonably be expected to be used to identify a natural person, all objective factors should be taken into account, such as the costs and time necessary for identification, taking into account the available technology at the time of processing.

and the technological development. It is stated in recital 26 of the

Data Protection Regulation.

It is not a prerequisite that the additional knowledge required to identify the individual is in the

possession of the data controller. It must be assessed whether

there are means available to the data controller that

can reasonably be expected to be utilized. The Court of Justice of the EU has clarified that this is not the case

if the identification of the data subject is prohibited

or practically unfeasible, for example, due to the fact

that identification would require a greater

investment of time, costs, and effort, such that

the risk of identification actually seems

insignificant.[4]

In this case, the assessment of whether a voice in

itself constitutes personal data must be made in light

of the fact that everyone will have access to the information, as it

will be published on the internet. This is because

the Alexandra Institute expects to publish

the dataset.

The current technology and the technological development seem to be of crucial

importance in this specific case for the assessment of whether the dataset

constitutes personal data, as

voice recognition tools have become

more common and accessible in recent years. It does not

therefore require access to special tools to conduct such analyses. Furthermore, it can be mentioned

that voice analysis is already being used

for identification, for example, in criminal cases, where it

is used to identify potential perpetrators and

other individuals based on their voice.

On social media platforms such as Instagram, TikTok, and

YouTube, hours of video clips are uploaded every day, featuring

people's voices. These videos are often accessible to a wide audience and may possibly be used as a basis for comparison when there is a desire for identification.

Moreover, the technological development, including the proliferation of artificial intelligence, has enabled the imitation of a person's voice for malicious purposes. There may therefore be an interest for a malicious actor to identify the person behind the audio recording in order to use the voice recording to carry out targeted IT security attacks (phishing) or fraud.

The dataset is also accompanied by metadata regarding gender, age, and approximate geographical location, which narrows down the pool of possible individuals behind the audio recording.

On this basis, the Data Protection Authority's assessment is that there are means that can reasonably be expected to be utilized to identify the individuals appearing in the dataset, and that the audio recordings must therefore be regarded as personal data.

Data protection rules therefore apply to the processing of the dataset, even after the dataset has been cleansed of any metadata.

3.3 Biometric Data

Finally, the inquiry raises considerations about whether the processing of the information in the relevant dataset will be covered by Article 9 of the Data Protection Regulation, including whether the voices in the dataset should be considered biometric data, and – if so – whether one of the exceptions in Article 9, paragraph 2, applies in this situation.

Article 4, number 14 of the Data Protection Regulation defines biometric data as personal data that, through specific technical processing related to the physical, physiological, or behavioral characteristics of a natural person, allows or confirms the unique identification of that person, e.g., facial images or fingerprint data.

Article 9, paragraph 1 of the Data Protection Regulation contains a general prohibition on processing special categories of personal data. This includes, among other things, biometric data for the purpose of uniquely

identifying a natural person.

Recital 51 further states:

“Processing of photographs should not be systematically considered to be processing of special categories of personal data, as they will only fall under the definition of biometric data when processed by a specific technical method that allows for the unique identification or authentication of a natural person.”

The Alexandra Institute has stated that the purpose of the project is to develop and improve Danish speech technology. In this context, the Institute will develop and publish a speech dataset that third parties can use to develop Danish speech technology.

Voices can, as mentioned, be unique due to differences in speech patterns, pitch, rhythm, pronunciation, language, and dialect. These characteristics can be used to differentiate one person's voice from another.

It is the Data Protection Authority's assessment that the information in the dataset may constitute biometric data. This depends on how the Alexandra Institute specifically expects to process the dataset, and this will be the case if the audio recordings are subjected to specific technical processing that allows for a unique identification of the person behind the voice.

However, it is the Data Protection Authority's assessment that the processing – even if it may involve biometric data – is not covered by the prohibition on processing in Article 9, paragraph 1 of the Data Protection Regulation. The Data Protection Authority particularly emphasizes that the Alexandra Institute's processing of the information does not occur with the aim of uniquely identifying a natural person.[5]

[1] C-252/21, Meta Platforms et al., paragraph 98.

[2] WP 136 – Opinion No. 4/2007 on the concept of personal data.

[3] The GDPR & Speech Data: Reflections of Legal

and Technology Communities, First Steps towards a Common Understanding, A. Nautsch et al. Electrical Engineering and Systems Science - Audio and Speech Processing, Computer Science - Computers and Society, 2019: <https://doi.org/10.48550/arXiv.1907.03458>

[4] C-582/14, Breyer, paragraph 46.

[5] Reference is also made to section 3.2 of the Data Protection Authority's opinion in the case with j.nr. 2018-211-01350, which can be found here: <https://www.datatilsynet.dk/afgoerelser/afgoerelser/2019/maj/vejledende-udtalelse-om-anvendelsen-af-fingeraftryk-til-brug-for-registrering-af-ansattes-komme-gaa-tider>